

Team Devdoots | Assignment #04 (Final Capstone): Advanced Interactive Portfolio & Swarm Robot Dashboard

Assigned By: Technical Project Manager (TPM)

Assignment Folder Name: Assignment_04_Capstone_Dashboard/

Branch Naming Format: user/yourname-assign4 (e.g., user/balram-assign4)

File Naming Format: Assignment_04_Capstone_Dashboard/yourname_dashboard.html & yourname_style.css

Commit Format: [HW-4] yourname completed capstone dashboard

Scope: Final HTML5 & Advanced CSS Capstone Project

1. Executive Summary & Objective

Yeh assignment Team Devdoots ke liye HTML5 aur CSS3 ka final capstone project hai. Iska main goal ek production-ready, structured, responsive aur modern single-page web dashboard banana hai. Is project me aapko apne personal member portfolio ko real-time Swarm Robot telemetry monitoring interface ke saath combine karna hai.

Is assignment se aapki semantic layout design, CSS Flexbox, CSS Grid, CSS root variables, animations, form layouting, aur official GitHub branch-and-PR workflow ki mastery check hogi.

Strict Policy on AI Usage & Code Generation:

Direct kisi AI tool se pura code generate karwaya hua copy-paste karna **strictly prohibited** hai. AI ka use aap sirf kisi specific CSS property ko samajhne, layout debugging, ya rare syntax confusion ke liye kar sakte ho. Code review ke time har team member ko apni line-by-line logic TPM ko explain karni hogi.

2. Project Specification: What to Build (Kya Banana Hai)

Section A: Sticky Header & Dynamic Navigation (Flexbox)

- Ek sticky top navigation bar (position: sticky) jo scroll karte waqt top par fixed rahe.
- Header me Team Brand (Team Devdoots), Member Avatar/Profile, Role Title (TPM / Engineer / Data Scientist), aur Navigation Links honge.
- Navigation links par smooth CSS hover animation effect hona chahiye (jaise expanding underline ya smooth color shift).

Section B: Member Profile & Tech Stack Matrix (CSS Grid)

- Ek multi-column card layout jo purely CSS Grid (e.g., display: grid) se bana ho.
- Cards me aapke core technical skills (Python, HTML/CSS, SQL, Git, Robotics) honge.
- Har card me skill title, proficiency level indicator, aur domain tag hoga.
- Mobile screens par ye grid media queries ya auto-fit grid properties ka use karke automatically single-column layout me convert hona chahiye.

Section C: Swarm Robot Telemetry & Live Status (Advanced Tables & Flexbox)

- Ek live operational telemetry table jisme system metrics dikhegi (e.g., Node ID, Battery Level, Operating Status, Signal Strength).
- Table me clean padding, custom header highlights, alternating row colors (zebra striping using :nth-child(even)), aur row hover highlights honge.
- System status badges me CSS @keyframes animation ka use karke ek pulsating live green dot indicator banana hai.

Section D: Interactive Task & Telemetry Feedback Form (CSS Variables)

- Ek modern form container jisme Text Inputs (Task Name), Dropdown Selects (Status), File Upload (Log Upload), aur Textareas (Comments) honge.
- Poore theme ke colors aur design elements global CSS Root Variables (:root) se control honge.
- Submit button me active aur hover states hone chahiye (e.g., smooth lift effect using transform: translateY(-2px)).

3. Implementation Guide: How to Build (Kaise Banana Hai)

Step 1: HTML Architecture & Structure

Apne assignment folder ke andar apni unique file banao (e.g., balram_dashboard.html). Tags ko properly nest karo aur overused raw <div> ke bajaye semantic tags (<header>, <main>, <section>, <footer>) use karo.

Step 2: Configure CSS Variables & Reset

Apni CSS file (e.g., balram_style.css) ki starting me global design tokens define karo:

```
:root {
  --bg-dark: #0f172a;
  --card-bg: #1e293b;
  --primary-blue: #2563eb;
  --accent-cyan: #38bdf8;
  --text-light: #f8fafc;
  --border-color: #334155;
  --success-green: #22c55e;
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: 'Segoe UI', system-ui, sans-serif;
}

body {
  background-color: var(--bg-dark);
  color: var(--text-light);
  line-height: 1.6;
}
```

Step 3: Build the Sticky Header using Flexbox

```
header {  
  position: sticky;  
  top: 0;  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  background-color: rgba(30, 41, 59, 0.95);  
  padding: 1rem 2rem;  
  border-bottom: 1px solid var(--border-color);  
  z-index: 1000;  
}
```

```
nav a {  
  color: var(--accent-cyan);  
  text-decoration: none;  
  margin-left: 1.5rem;  
  position: relative;  
  padding-bottom: 4px;  
}
```

```
nav a::after {  
  content: " ";  
  position: absolute;  
  width: 0;  
  height: 2px;  
  bottom: 0;  
  left: 0;  
  background-color: var(--accent-cyan);  
  transition: width 0.3s ease;  
}
```

```
nav a:hover::after {  
  width: 100%;  
}
```

Step 4: Implement Responsive Cards using CSS Grid

```
.grid-container {  
  display: grid;
```

```
    grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
    gap: 1.5rem;
    margin-top: 1rem;
}
```

```
.card {
    background-color: var(--card-bg);
    padding: 1.5rem;
    border-radius: 8px;
    border: 1px solid var(--border-color);
    transition: transform 0.2s ease, box-shadow 0.2s ease;
}
```

```
.card:hover {
    transform: translateY(-4px);
    box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.3);
}
```

Step 5: Add Pulsating Status Badge Animation

```
@keyframes pulse {
    0% { transform: scale(0.95); opacity: 0.7; }
    50% { transform: scale(1.1); opacity: 1; }
    100% { transform: scale(0.95); opacity: 0.7; }
}
```

```
.live-badge {
    display: inline-block;
    width: 10px;
    height: 10px;
    background-color: var(--success-green);
    border-radius: 50%;
    margin-right: 6px;
    animation: pulse 2s infinite ease-in-out;
}
```

4. Official GitHub Workflow & Submission Rules (The Workspace Method)

Why This Workflow?

Chunki hum sabhi ek hi repository par kaam kar rahe hain, agar koi direct main branch me code push karega toh Merge Conflicts aayenge aur repository kharab ho jayegi. Isliye sabhi team members ko strictly neech diye gaye steps follow karne honge.

Step-by-Step Submission Guide:

Step 1: Latest Code Pull Karo

Sabse pehle local main branch ko update karo taaki TPM dwara banaya gaya naya assignment folder aapke paas aa jaye:

```
git checkout main
```

```
git pull origin main
```

Step 2: Apne Naam Ki Personal Branch Banao

Direct main me koi kaam nahi karega. Apne naam ki branch banao format: user/name-task

```
git checkout -b user/balram-assign4
```

Step 3: Golden Rule - Personal File Naming

Folder Assignment_04_Capstone_Dashboard/ ke andar koi bhi common file (jaise index.html) nahi banayega. Aapko apne naam se file banani hai taaki kisi ka code takraye nahi.

Example: Assignment_04_Capstone_Dashboard/balram_dashboard.html aur balram_style.css

Step 4: Committing Your Code (Proper Format)

Kaam poora hone par proper commit message format me commit karo: [HW-Number] name completed task

```
git add .
```

```
git commit -m "[HW-4] balram completed capstone dashboard"
```

Step 5: Push Branch & Create Pull Request (PR)

Branch ko push karo aur GitHub par jaakar main branch ke against Pull Request (PR) create karo:

```
git push origin user/balram-assign4
```

Summary Command Line Flow:

1. `git pull origin main`
2. `git checkout -b user/yourname-assign4`
3. Create file: `Assignment_04_Capstone_Dashboard/yourname_dashboard.html`
4. `git add .`
5. `git commit -m "[HW-4] yourname completed capstone dashboard"`
6. `git push origin user/yourname-assign4`
7. Go to GitHub and create a Pull Request (PR).

5. PR Review & Acceptance Checklist (TPM Verification)

Category	Requirement Criteria	Status
File Naming	File assignment folder ke andar yourname_dashboard.html format me hai	Required
Branch Standard	Branch name strictly user/yourname-assign4 format me hai	Required

Category	Requirement Criteria	Status
Commit Standard	Commit message format [HW-4] name completed task hai	Required
Architecture	Strict use of semantic HTML5 tags (<header>, <main>, <section>, <footer>)	Required
CSS Standards	No inline styling; clean external stylesheet using CSS Variables (:root)	Required
Layout Engine	CSS Grid used for cards; Flexbox used for header/nav and status indicators	Required
Responsiveness	Layout adapts smoothly across desktop, tablet, and mobile breakpoints	Required

Note: PR review ke baad agar koi galti hogi toh TPM comments me changes bataenge. Code 100% sahi hone par hi main branch me merge kiya jayega.